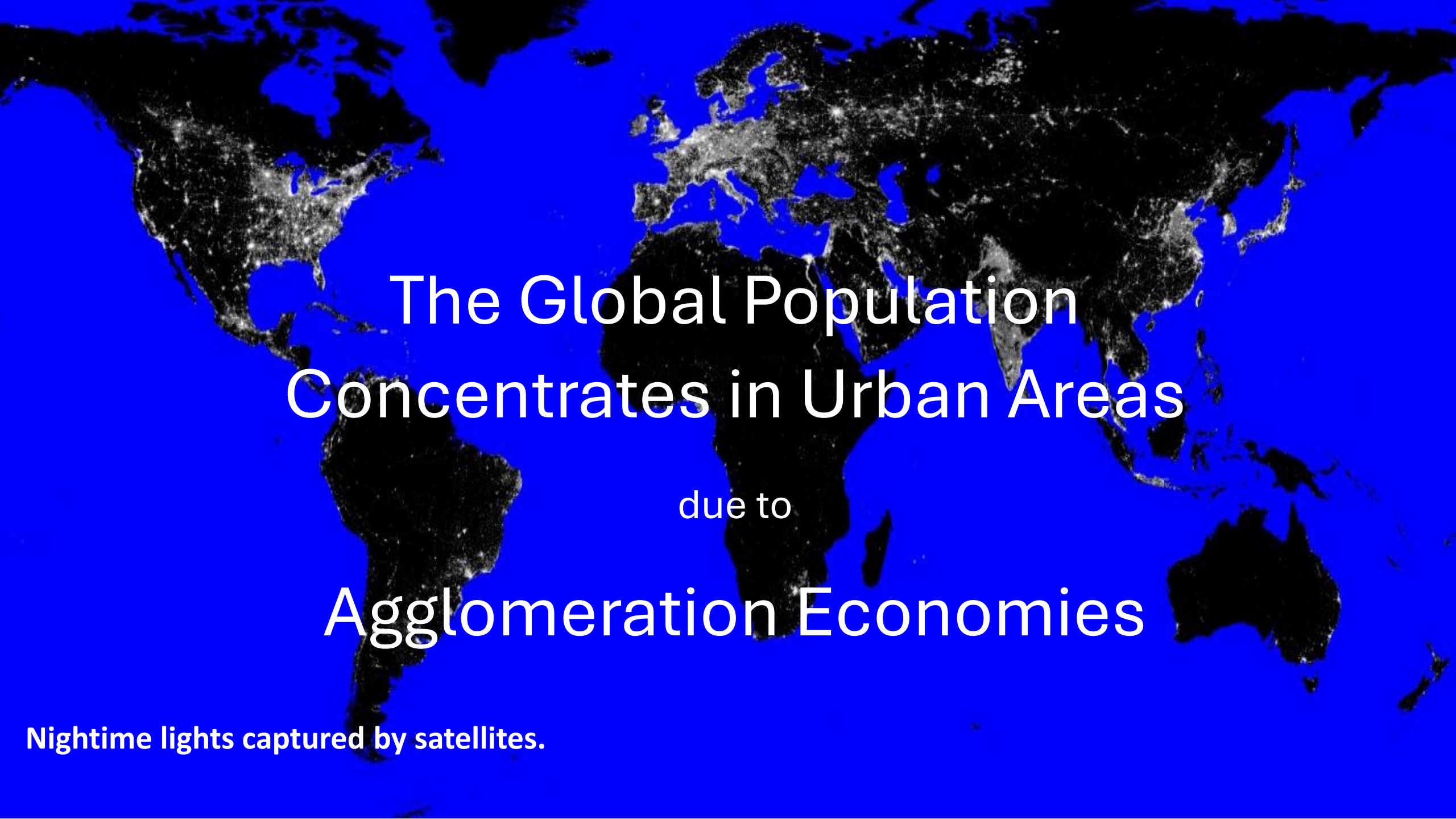


The development of cities and regions 1: agglomeration, innovation, and technological development

Balázs Lengyel

Economic Complexity, Class 3 – Sept 30 2026

Social Data Science, Corvinus University of Budapest

A world map with a black background, where landmasses are outlined in white. The map is filled with numerous small, bright white dots and larger, irregular white patches, representing nighttime lights captured by satellites. These lights are most concentrated in North America, Europe, and East Asia, with significant clusters in the eastern United States, Western Europe, and East Asia. Other smaller clusters are visible in South America, Africa, and Australia. The oceans are represented by the black background.

The Global Population Concentrates in Urban Areas

due to

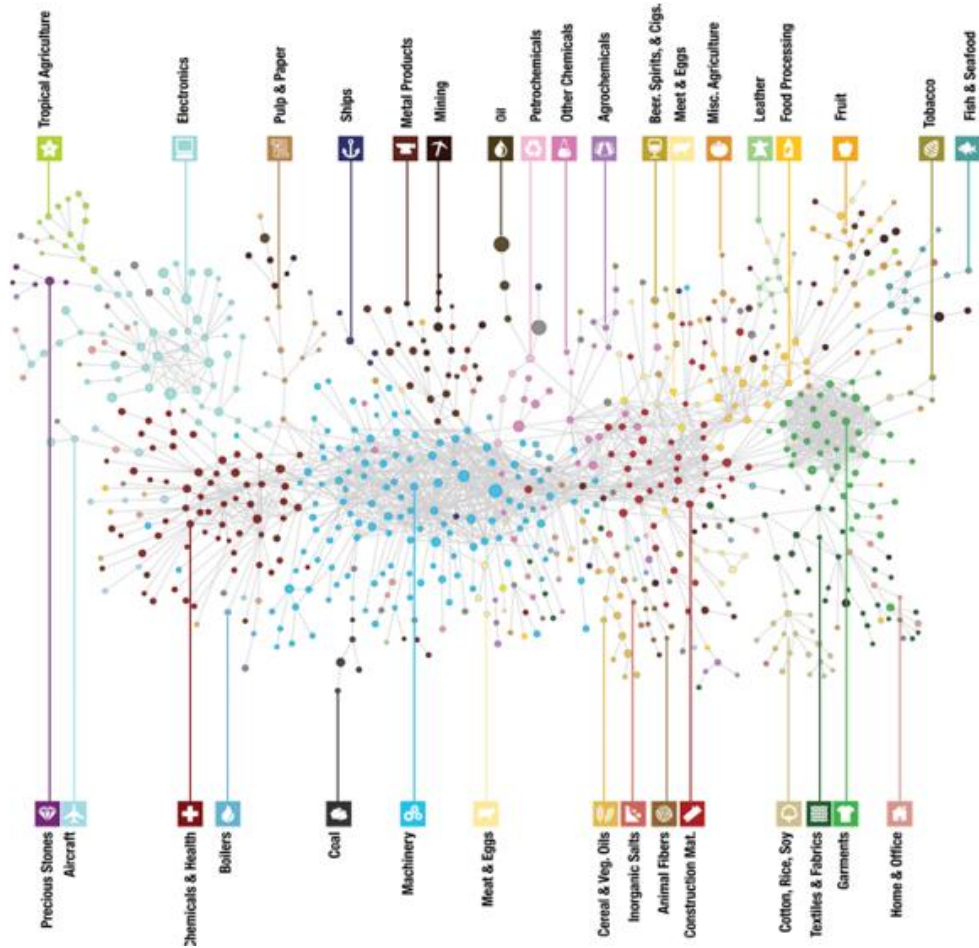
Agglomeration Economies

Nighttime lights captured by satellites.

1992



Product space: assumptions and messages



- Capabilities/knowledge
- Spatial specialization

Today

1. What is knowledge/innovation in economic development?
2. Location theory: costs and markets
3. Centrum and hierarchy
4. Agglomeration economies
5. Concentration of knowledge and innovation

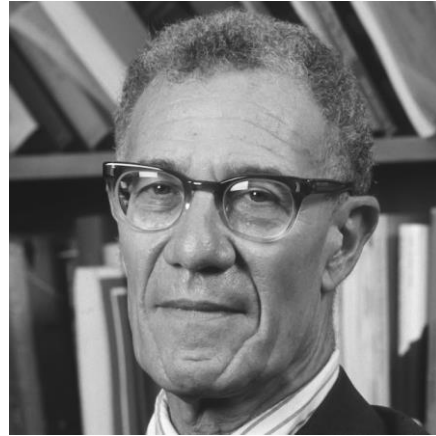
1. Knowledge in economics

Technological development and growth

Robert Solow (MIT)

Technology in growth

Nobel prize in 1987



$$Y = F(K, AL) = K^\alpha (AL)^{1-\alpha}$$

$$A(t) = A(0)e^{gt}$$

g is exogenous technological development

Paul Romer (Princeton)

Endogenous growth

Nobel prize in 2018



$$L = L_A + L_Y$$

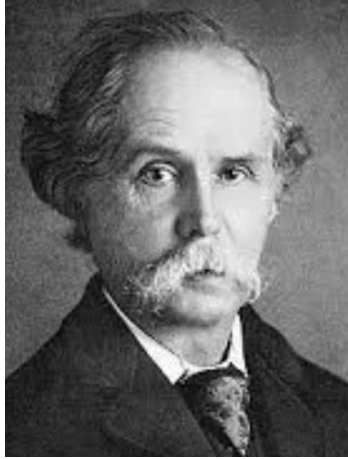
$$\dot{A} = \gamma L_A^\lambda A^\phi$$

$$Y = (AL_Y)^{1-\alpha} K^\alpha$$

Access to knowledge

Alfred Marshall
(Cambridge)

Knowledge is in the air



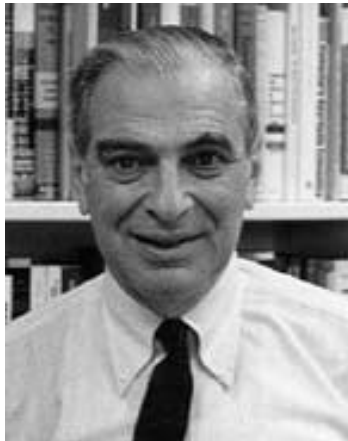
Matthew Jackson
(Stanford)

Learning in networks

Kenneth Arrow (Harvard)

Learning by doing

Nobel prize in 1987



Daron Acemoglu (MIT)

Nobel prize in 2024

Claude Shannon (MIT)

Information theory

A Mathematical Theory of Communication

By C. E. SHANNON

9. THE FUNDAMENTAL THEOREM FOR A NOISELESS CHANNEL

We will now justify our interpretation of H as the rate of generating information by proving that H determines the channel capacity required with most efficient coding.

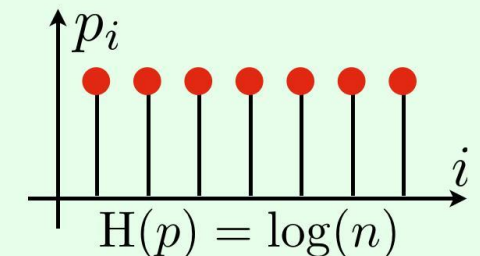
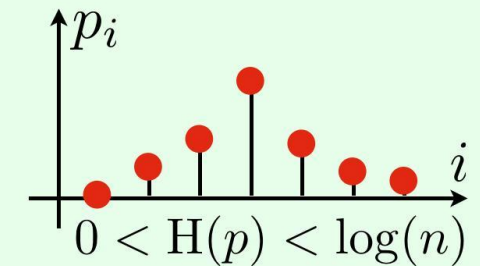
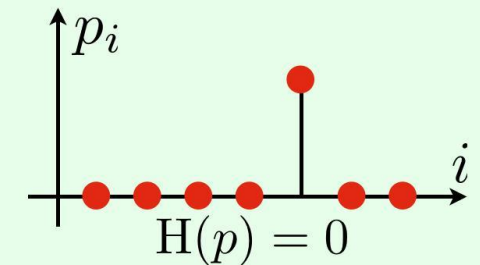
Theorem 9: Let a source have entropy H (bits per symbol) and a channel have a capacity C (bits per second). Then it is possible to encode the output of the source in such a way as to transmit at the average rate $\frac{C}{H} - \epsilon$ symbols per second over the channel where ϵ is arbitrarily small. It is not possible to transmit at an average rate greater than $\frac{C}{H}$.

Information theory is about information.
Information theory explains information.

Information theory quantifies uncertainty,
compresses data, and underlies modern
cryptography and machine learning.



$$H(p) \stackrel{\text{def.}}{=} - \sum_i p_i \log(p_i)$$



Michael Polanyi (Manchester)

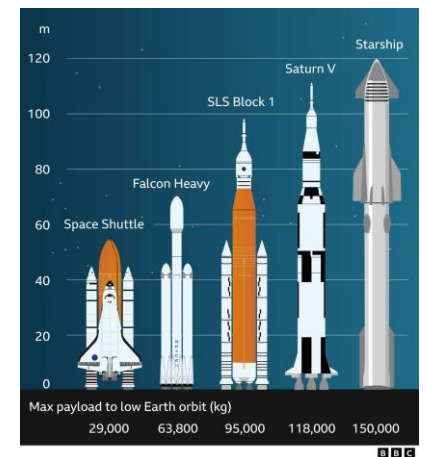
Tacit knowledge



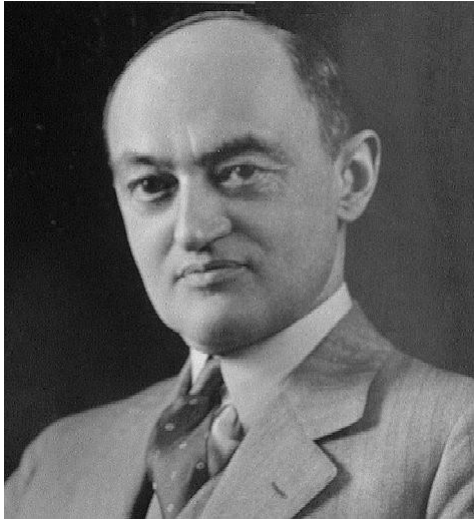
„We know more than we can tell.”

- **Personal and Experiential:** rooted in individual's experiences and shaped by specific context
- **Intuitive:** feelings, insights, hard to explain
- **Difficult to Document:** cannot easily be written down, spoken, or put into a standard procedure
- **Acquired Through Practice:** learned through doing, observing, and interacting
- **"Know-How" vs. "Know-That":** practical skill or understanding of how to do something, in contrast to explicit knowledge's focus on facts ("know-that")

Technological development



Evolutionary economics



Joseph Schumpeter (Harvard)

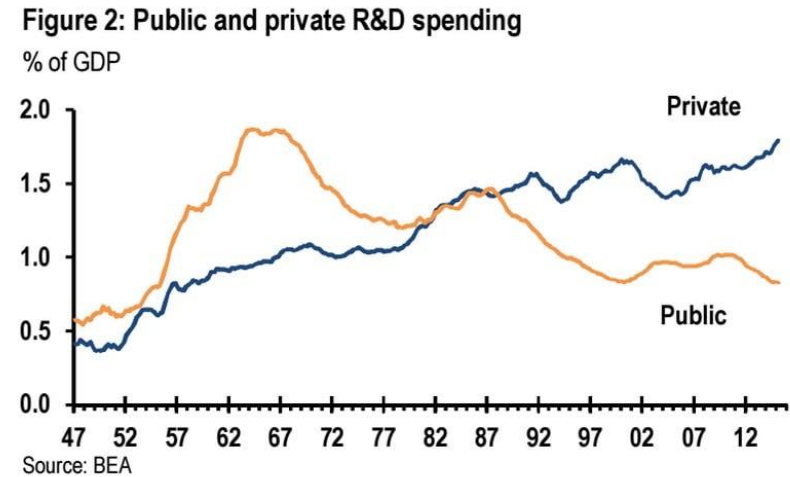
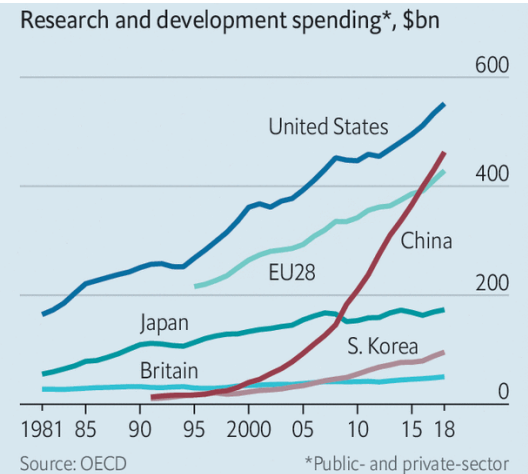
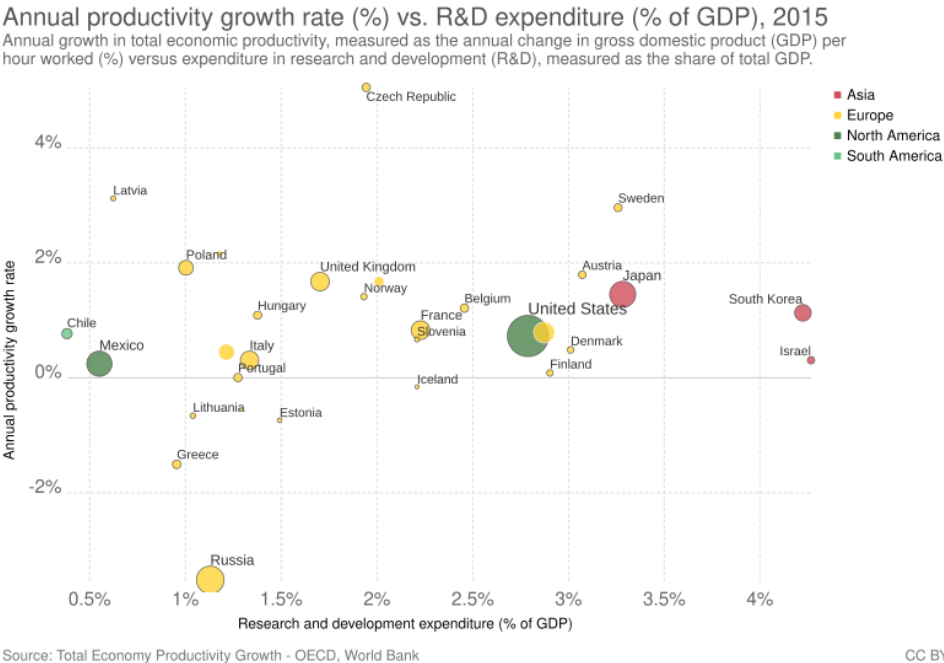
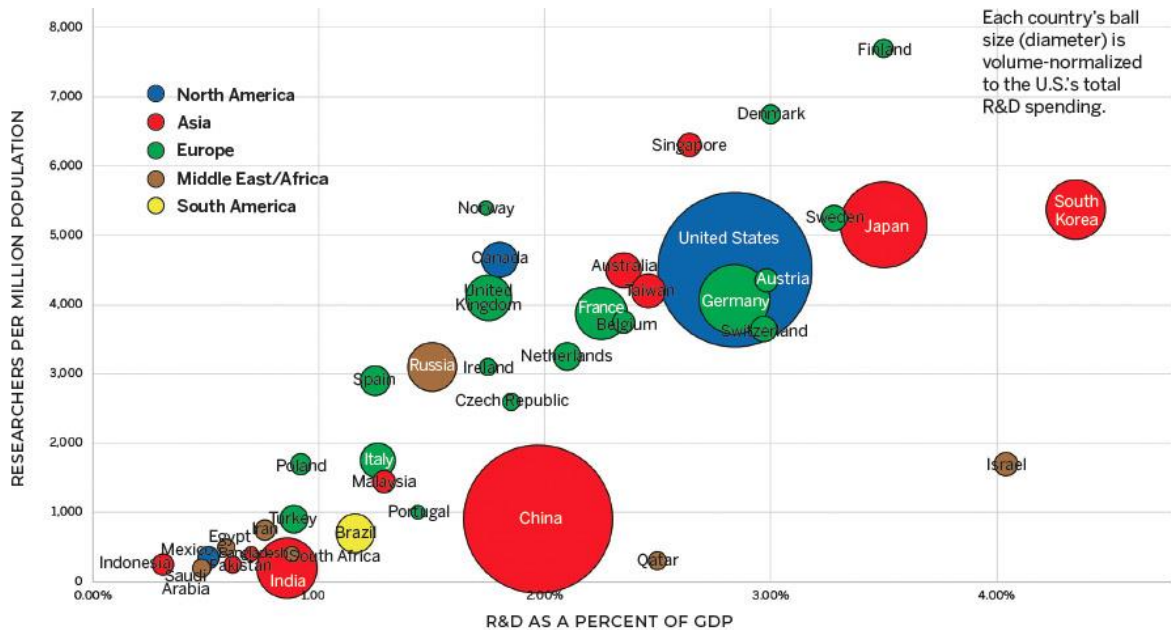
- Economic development is driven by innovation and technological development
- Creative destruction: old technologies are replaced by new one
- The Entrepreneur is the major driver of development

Concept in Biology	Schumpeterian Analogy
Mutation	Innovation
Natural selection	Market competition
Fitness	Profitability, survival
Extinction	Bankruptcy, firm failure
Speciation	Industry creation or transformation

Small versus Big



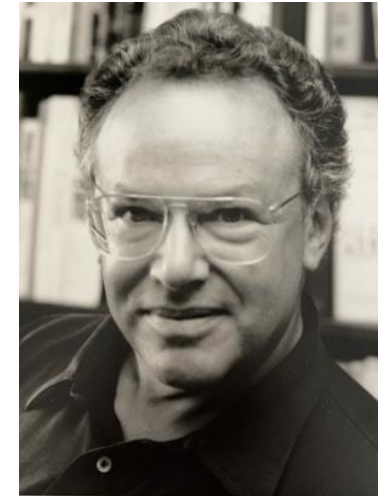
R&D facts



Path dependence, lock-in

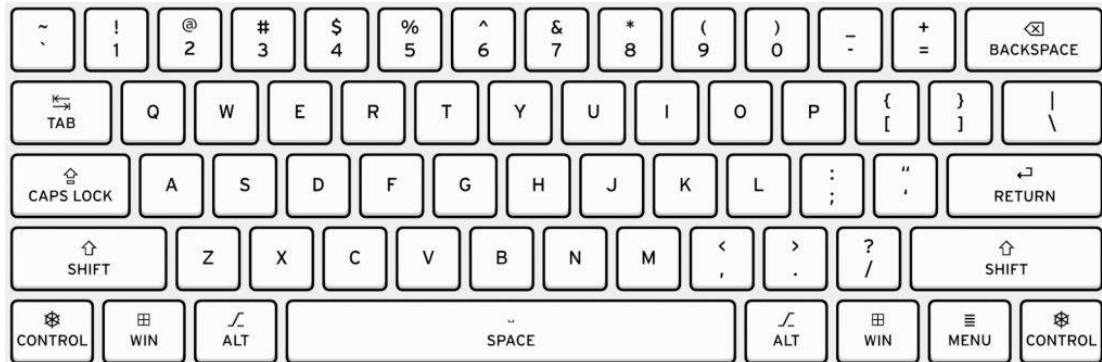


Paul David (Stanford)



Key Mechanisms:

- **Historical accident:** QWERTY got early traction through deals with Remington, the dominant typewriter producer.
- **Learning effects:** People and institutions invested time and effort to learn QWERTY.
- **Network effects:** The more people used QWERTY, the more valuable it became (shared training, manuals, etc.)
- **Switching costs:** Moving to a new layout (like Dvorak) would require retraining, software changes, and user resistance — even if it's better.
- Result: **QWERTY is "locked in"** — not because it's the best, but because the **costs of switching** and the **advantages of compatibility** outweigh the benefits of alternatives.



The History of INNOVATION CYCLES

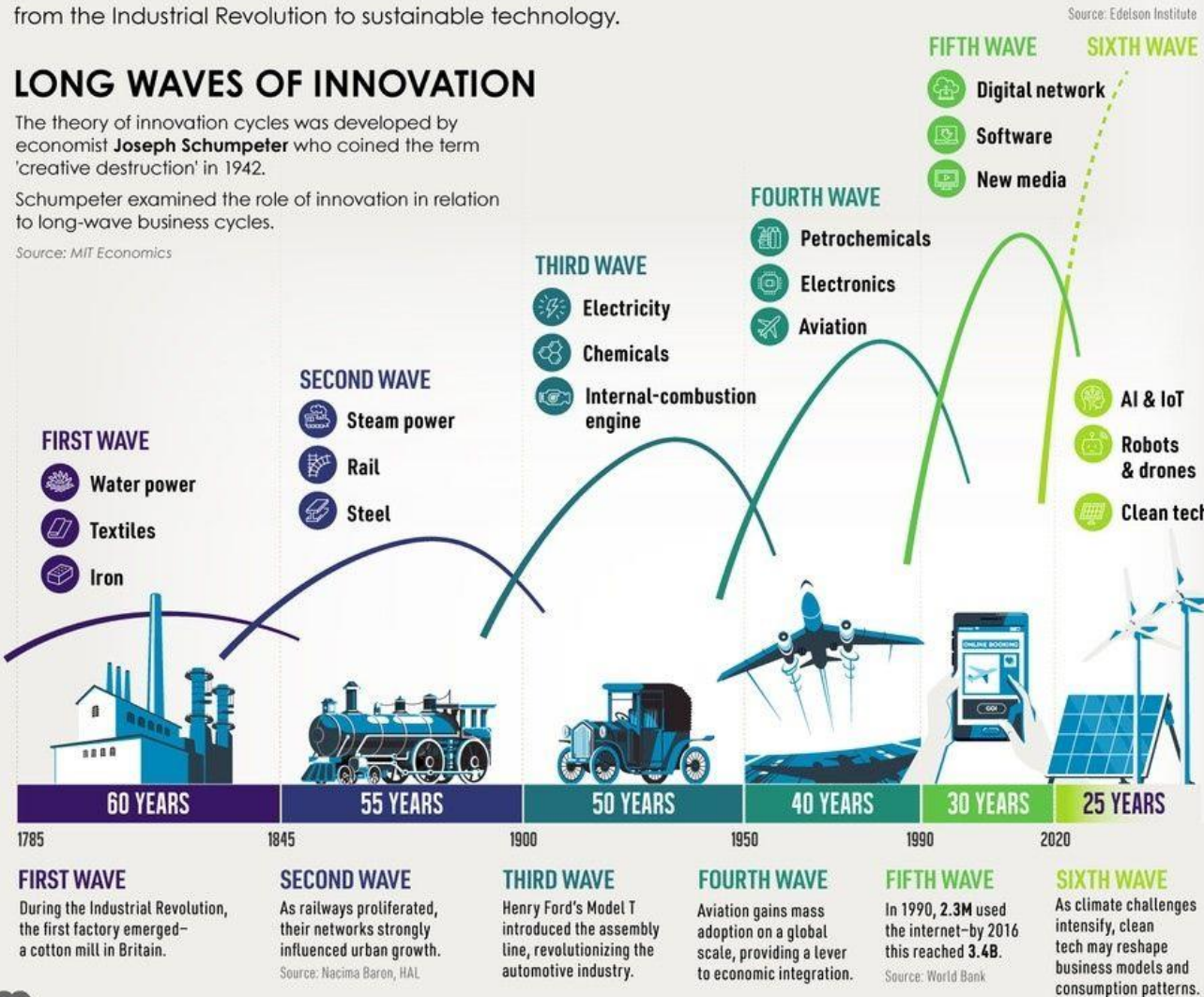
Below, we show waves of innovation across 250 years, from the Industrial Revolution to sustainable technology.

LONG WAVES OF INNOVATION

The theory of innovation cycles was developed by economist **Joseph Schumpeter** who coined the term 'creative destruction' in 1942.

Schumpeter examined the role of innovation in relation to long-wave business cycles.

Source: MIT Economics



Cycles of general purpose technologies (Perez 2002)

Mechanism

Technological paradigm

Financial speculation

Overinvestment

Crisis and correction

Institutional adaptation

Saturation and stagnation

Social pressure

Role

Sets the direction for innovation

Accelerates diffusion, but causes bubbles

Builds infrastructure, though inefficiently

Aligns tech and society post-bubble

Enables full deployment of GPT

Signals the end of a cycle

Drives the shift to the next paradigm

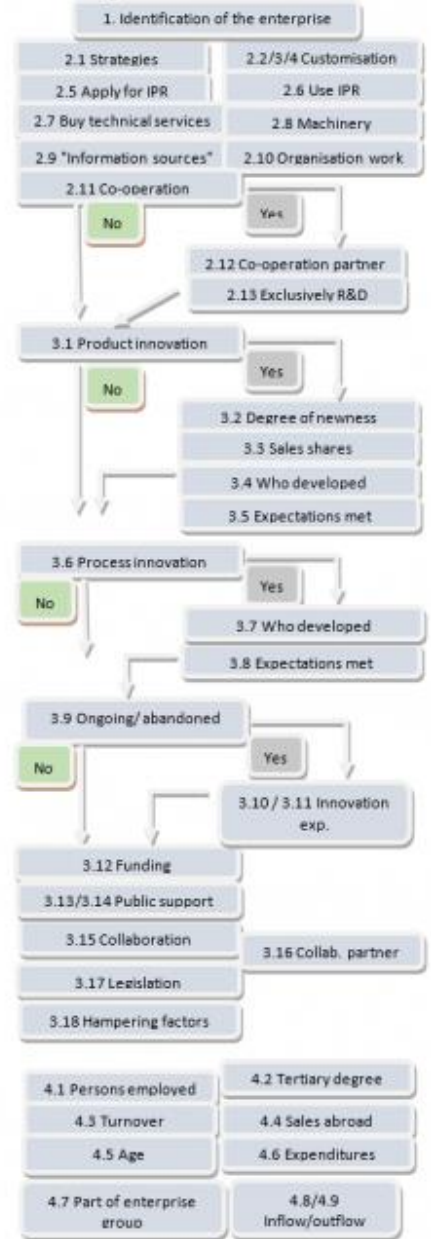
Measurement and recent research on
innovation

Measurements of surveys

Flow of questions in CIS 2016



Flow of questions in CIS 2018



Measurement by statistical offices



 Log in

 English

 Search

 Information

Eurostat is gradually releasing its metadata with a new design and layout. During this transition time, there will be a mix between metadata in the old and new look. To find out more, read our [news article](#).



Home

Data ▾

News ▾

Publications ▾

About us ▾

Contact us ▾

Help ▾

Home > Data > Statistical themes > Science, technology, and innovation > Database

SCIENCE, TECHNOLOGY, AND INNOVATION

Database

Overview

Database

Visualisations

Publications

Information on data

Methodology

Quality

Legislation

Selected datasets

 Science and technology (t_scitech)

 Research and development (R&D) (t_rd) 

 High-tech industry and knowledge-intensive services (t_htec) 

 Human Resources in Science & Technology (t_hrst) 

Detailed datasets

 Science and technology (scitech)

 Research and development (R&D) (rd)  [\(Information note\)](#)

 Community innovation survey (inn) [\(Information note\)](#)

 High-tech industry and knowledge-intensive services (htec) 

 Human Resources in Science & Technology (hrst) 

 Intellectual property rights - Patents (pat)

Measurement from documents

(19)

(11)**EP 3 155 128 B1**

Publication number

(12)**EUROPEAN PATENT SPECIFICATION**

(45) Date of publication and mention of the grant of the patent:
15.05.2019 Bulletin 2019/20

(51) Int. CL.:
C12Q 1/68 (2018.01) A61K 48/00 (2006.01)
C12N 15/11 (2006.01)

Patent classes

(21) Application number: **15732984.8**

(22) Date of filing: **10.06.2015**

(86) International application number:
PCT/NL2015/050420

(87) International publication number:
WO 2015/190921 (17.12.2015 Gazette 2015/50)

(54) **METHODS FOR CHARACTERIZING ALTERNATIVELY OR ABERRANTLY SPLICED MRNA ISOFORMS**
VERFAHREN ZUR CHARAKTERISIERUNG ALTERNATIV ODER ANOMAL GESPLEISSTER MRNA-ISOFORMEN
PROCÉDÉS DE CARACTÉRISATION D'ISOFORMES D'ARNM ÉPISSÉS DE MANIÈRE DIFFÉRENTE OU ABERRANTE

(84) Designated Contracting States:
AL AT BE BG CH CY CZ DE DK EE ES FI FR GB GR HR HU IE IS IT LI LT LU LV MC MK MT NL NO PL PT RO RS SE SI SK SM TR

(30) Priority: **10.06.2014 PCT/NL2014/050375**
04.09.2014 EP 14183623

(43) Date of publication of application:
19.04.2017 Bulletin 2017/16

(73) Proprietor: **Erasmus University Medical Center Rotterdam**
3015 GE Rotterdam (NL)

(72) Inventors:
• **BERGSMA, Atze Jacobus**
NL-3015 GE Rotterdam (NL)
• **VAN DER WAL, Erik**
NL-3015 GE Rotterdam (NL)
• **PIJNAPPEL, Wilhelmus Wenceslaus Matthias**
NL-3015 GE Rotterdam (NL)
• **VAN DER PLOEG, Antje Tjitske**
NL-3015 GE Rotterdam (NL)
• **REUSER, Arnoldus**
NL-3015 GE Rotterdam (NL)

(74) Representative: **V.O.**
P.O. Box 87930
Carnegieplein 5
2508 DH Den Haag (NL)

(56) References cited:
WO-A1-2007/135105 WO-A2-01/77380
WO-A2-2013/043878

• **OHNUMA S ET AL:** "Cancer-associated splicing variants of the CDCA1 and MSMB genes expressed in cancer cell lines and surgically resected gastric cancer tissues", **SURGERY, MOSBY, INC, US, vol. 145, no. 1, 1 January 2009 (2009-01-01), pages 57-68, XP025837299, ISSN: 0039-6060, DOI: 10.1016/J.SURG.2008.08.010 [retrieved on 2008-12-27]**
• **ROB W J COLLIN ET AL:** "Mid-frequency DFNA8/12 hearing loss caused by a synonymous TECTA mutation that affects an exonic splice enhancer", **EUROPEAN JOURNAL OF HUMAN GENETICS, vol. 16, no. 12, 25 June 2008 (2008-06-25), pages 1430-1436, XP055140967, ISSN: 1018-4813, DOI: 10.1038/ejhg.2008.110**
• **T. YU ET AL:** "Using a minigene approach to characterize a novel splice site mutation in human F7 gene causing inherited factor VII deficiency in a Chinese pedigree", **HAEMOPHILIA, vol. 15, no. 6, 1 November 2009 (2009-11-01), pages 1262-1266, XP055140970, ISSN: 1351-8216, DOI: 10.1111/j.1365-2516.2009.02064.x**

EP 3 155 128 B1

Note: Within nine months of the publication of the mention of the grant of the European patent in the European Patent Bulletin, any person may give notice to the European Patent Office of opposition to that patent, in accordance with the Implementing Regulations. Notice of opposition shall not be deemed to have been filed until the opposition fee has been paid. (Art. 99(1) European Patent Convention).

Printed by Jouve, 75001 PARIS (FR)

(Cont. next page)

Date of publication and grant of the patent

Date of filing

Countries where the patent has been filed

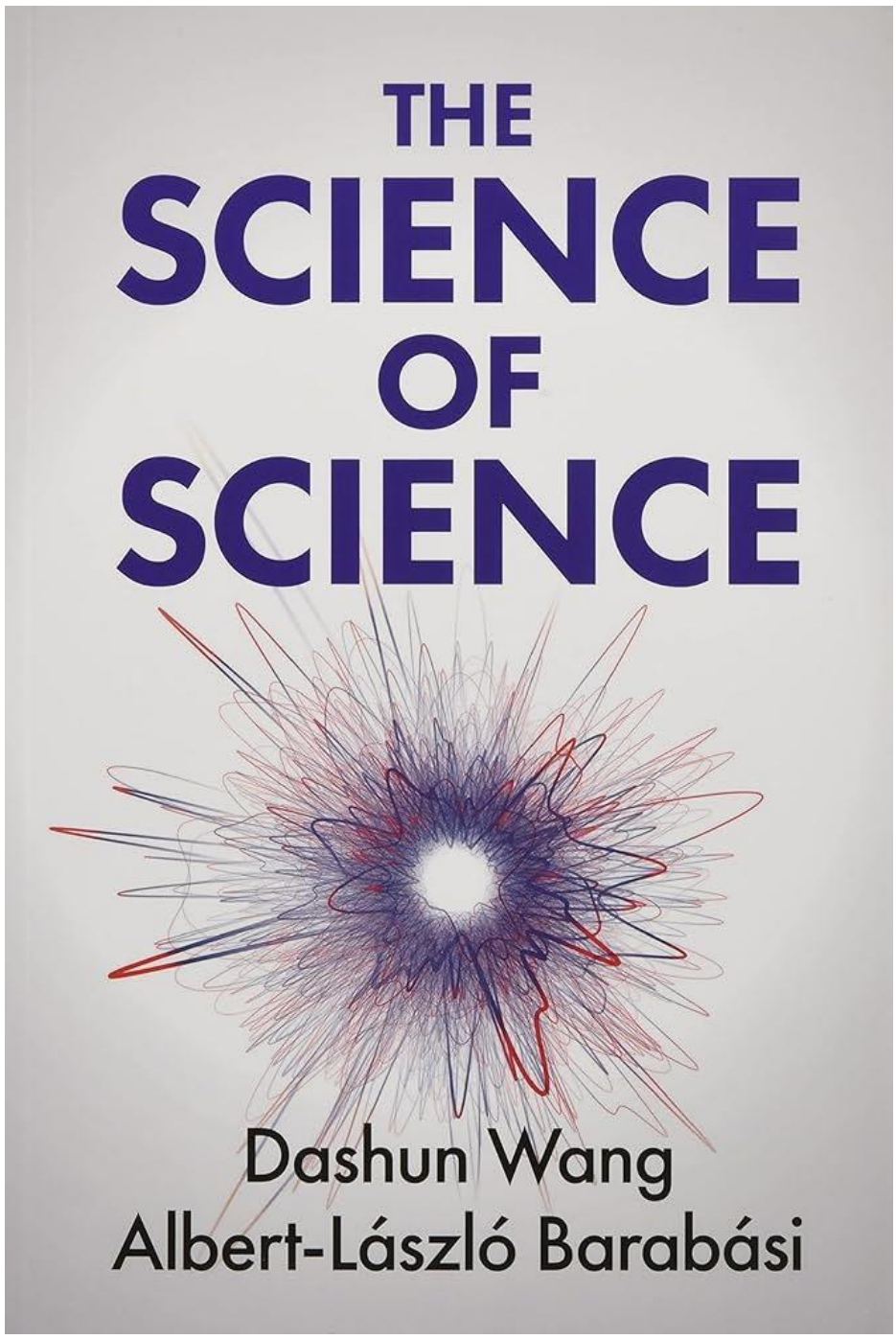
Priority date

Publication date

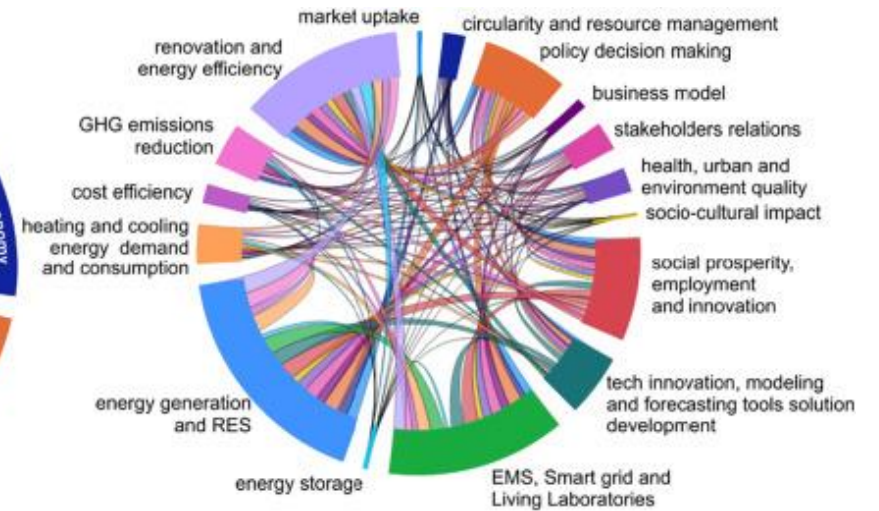
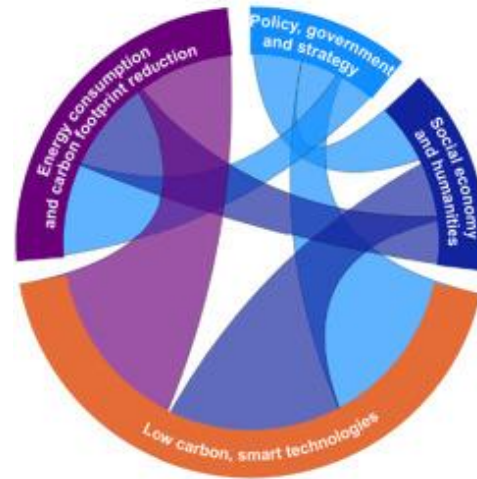
Applicant

Inventors

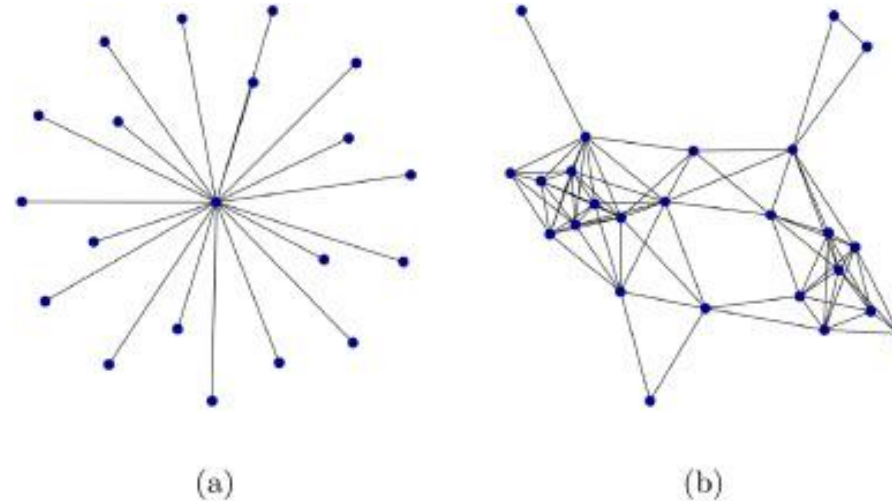
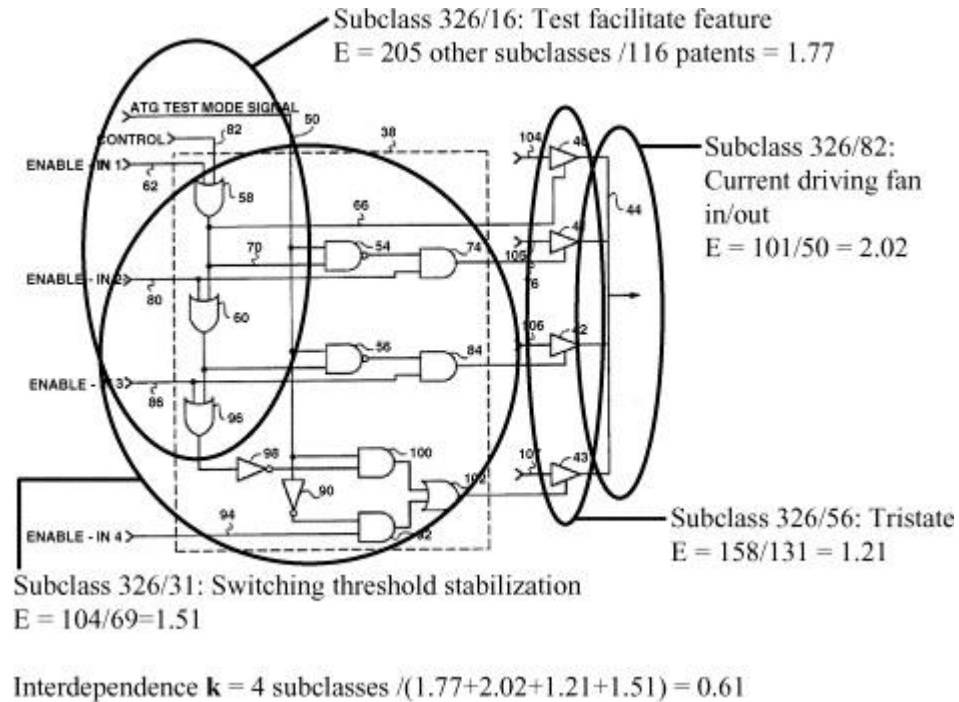
Representative



Research grants



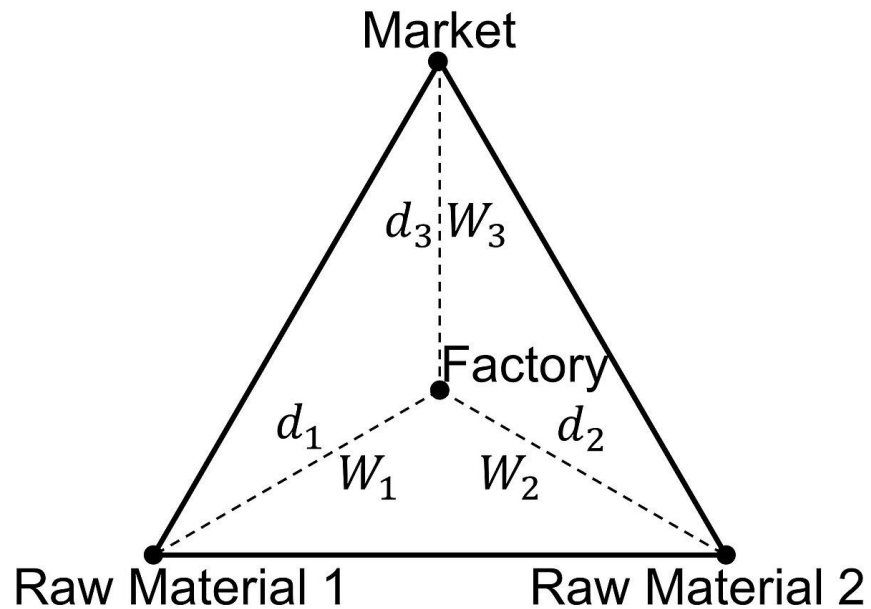
Combination of knowledge as measurement of knowledge complexity



Sorenson et al. 2006, Research Policy; Mewes and Broekel 2002, Research Policy

2. Location theory

Alfred Weber's location triangle



Looking for the minimal transportation cost location

$$TC = W_1 d_1 + W_2 d_2 + W_3 d_3$$

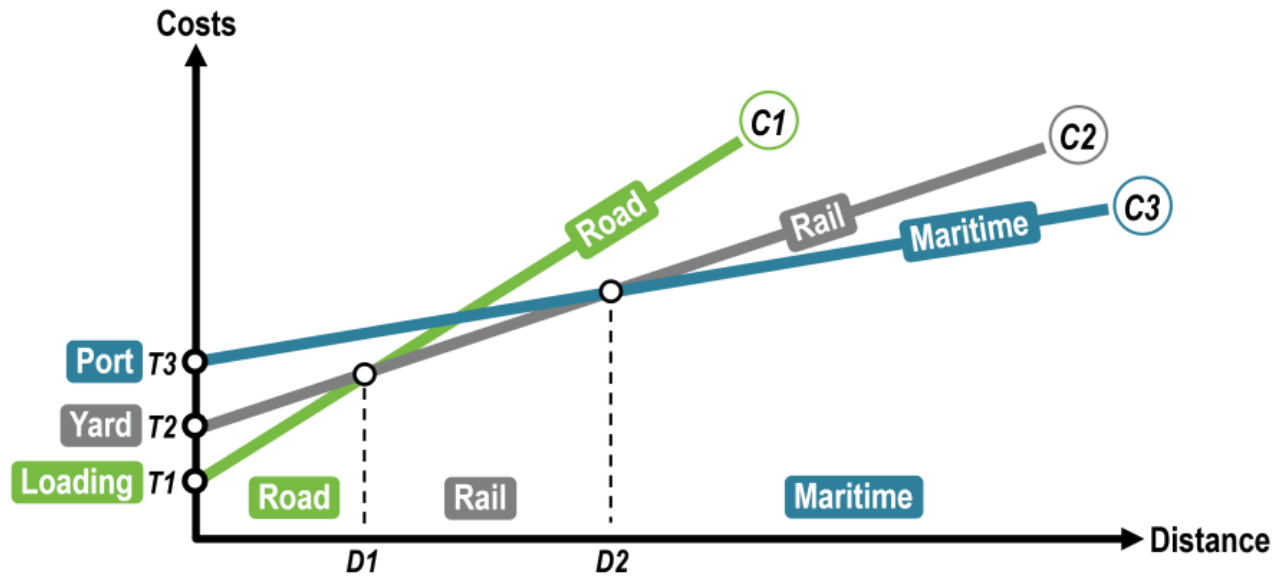
(W: weights)
(d: distance)

Ha

$$W_3 d_3 > W_1 d_1 + W_2 d_2 \Rightarrow \text{towards market}$$

$$W_3 d_3 < W_1 d_1 + W_2 d_2 \Rightarrow \text{towards raw materials}$$

Lösch: cost space and markets



Final cost is the sum of

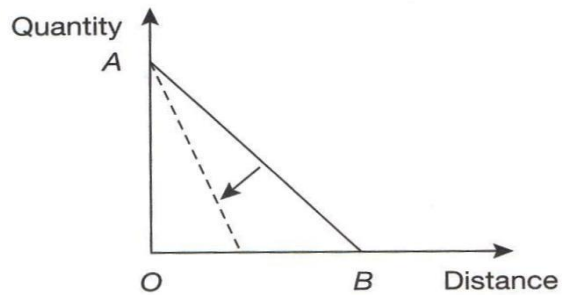
1. Free on Board (FOB):
production costs + loading
costs

2. Transportation costs to
markets

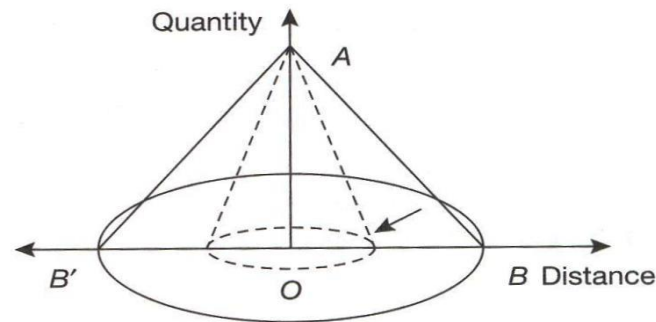
Lösch: spatial markets

(a) Increase in transportation costs

(a1) linear market

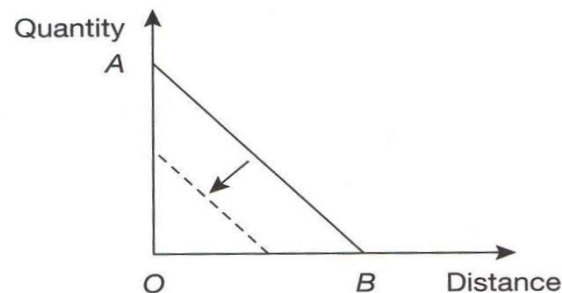


(a2) circular market (volume of the Lösch cone)

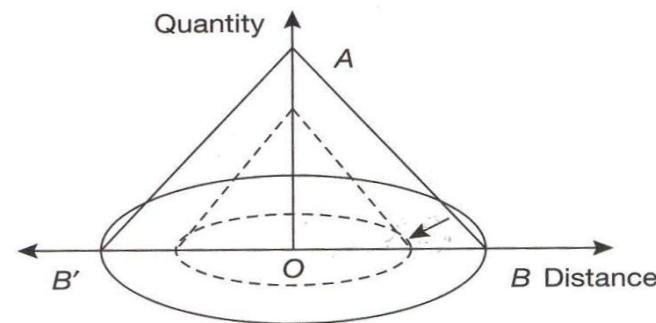


(b) Increase in the price of the good

(b1) linear market

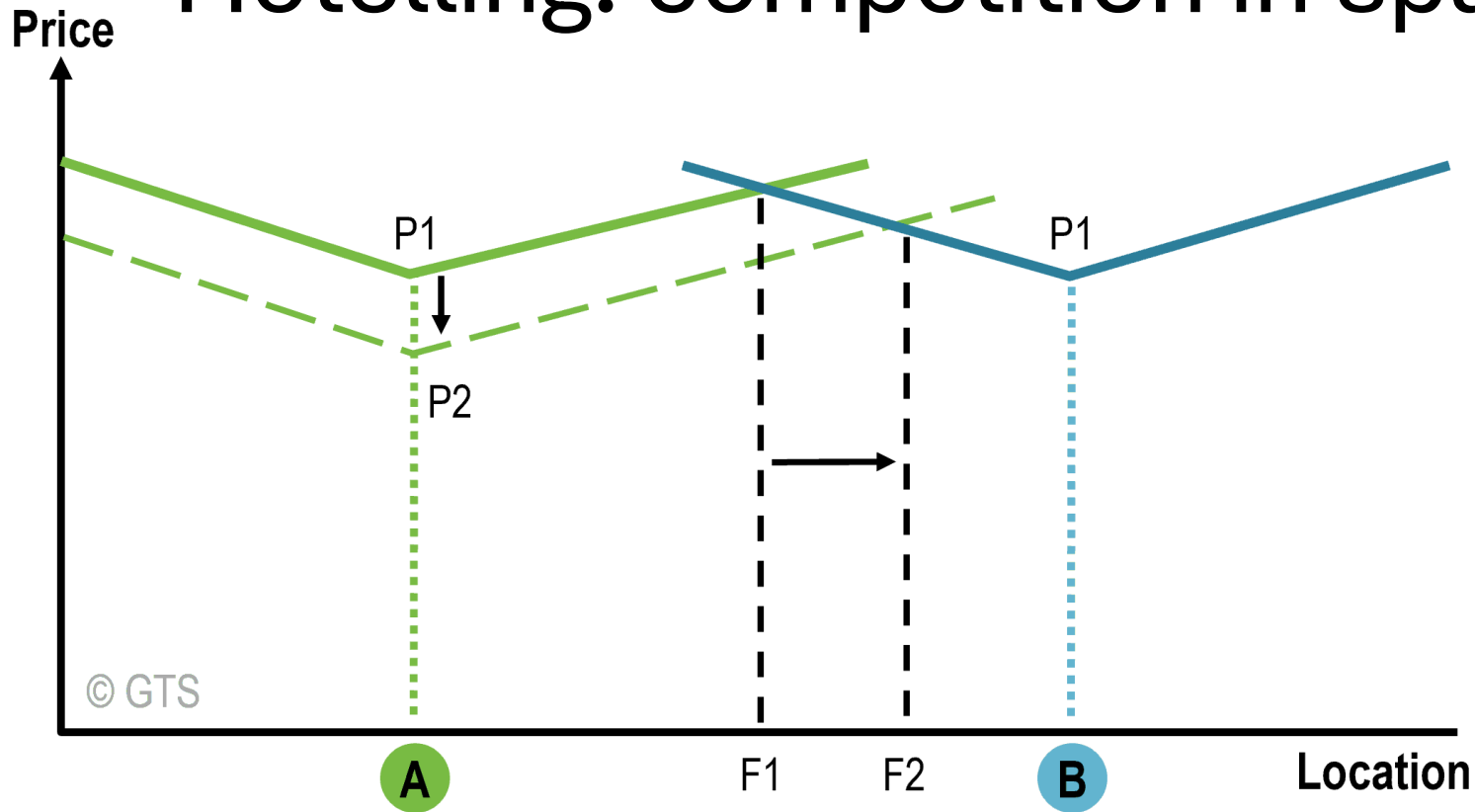


(b2) circular market (volume of the Lösch cone)

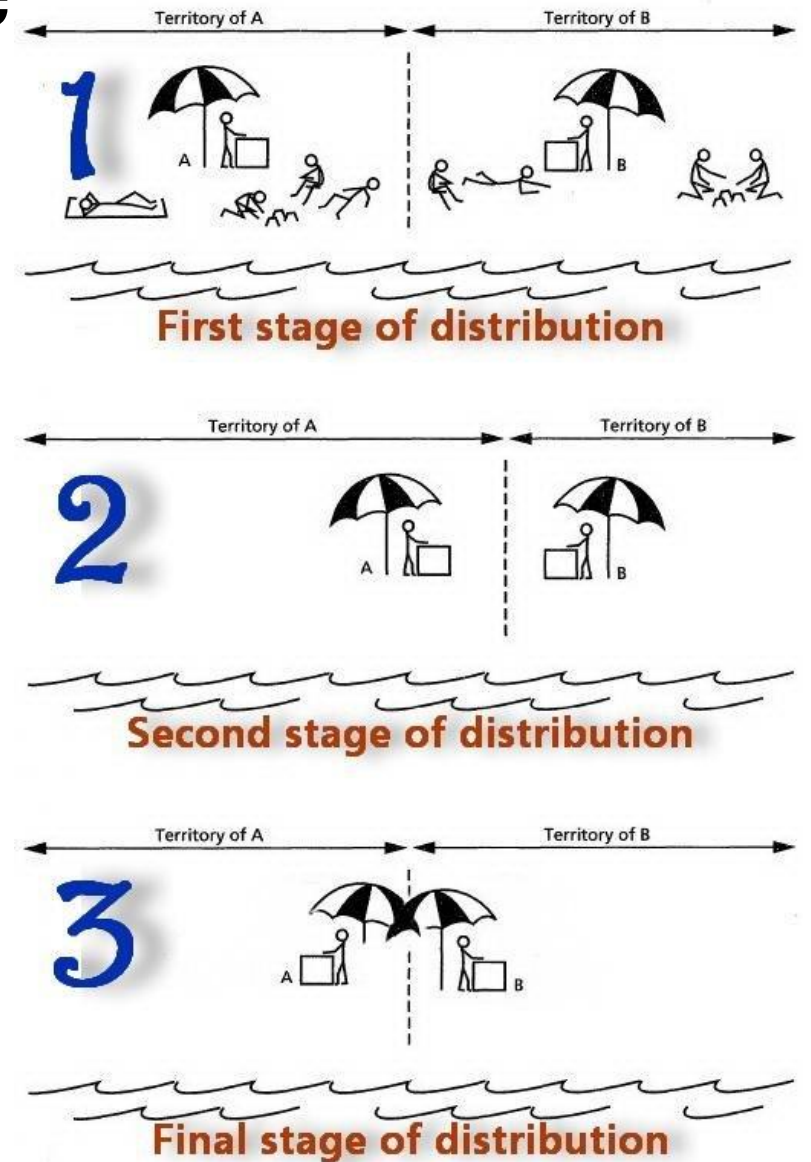


Increasing transportation costs and increasing product price leads to shrinking market size

Hotelling: competition in space



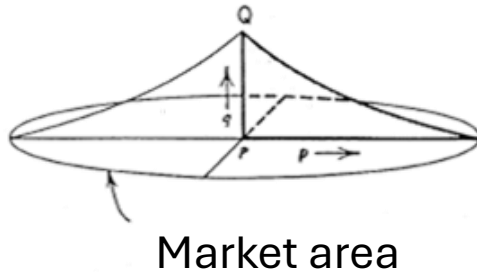
Firms co-locate to maximize market share



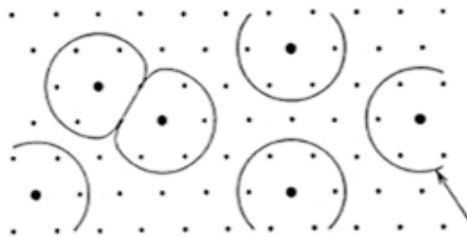
3. Centrum and hierarchy

Christaller: how do centers emerge?

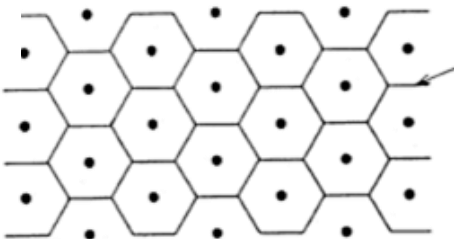
Phase 1



Phase 2



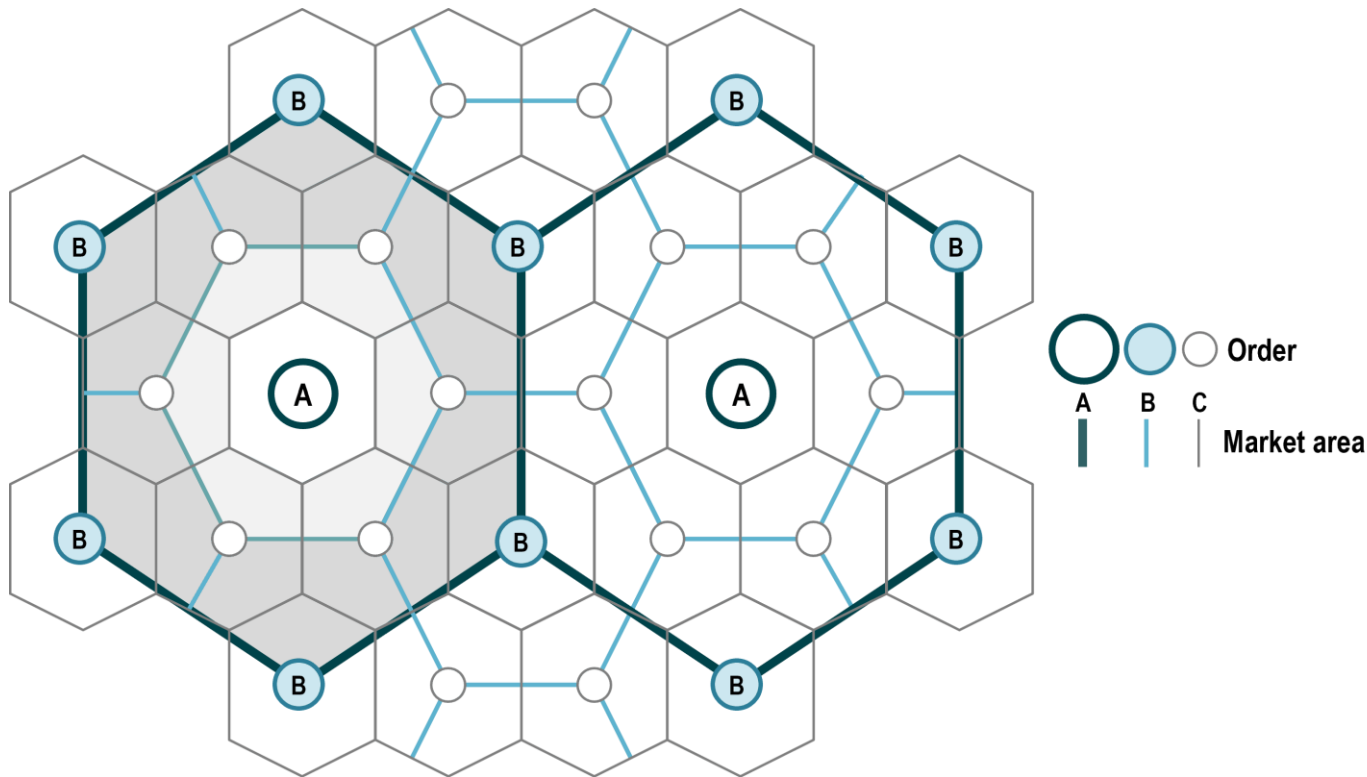
Phase 3



The Three Phases of Market Area Formation in a Single Industry:

1. **Single producer of goods** (natural market area)
2. **New firms enter** in the hope of earning extra profit, until supply covers the entire area. In overlapping zones, firms **compete** with one another.
3. A **long-term equilibrium** is established: all demand is satisfied, and natural market areas take on a **hexagonal shape**.

Christaller: central place theory



Derivation of the Urban Hierarchy:

- The **range** of services differ
- **Higher-order services** are only provided in **urban centers**
- **Higher-order centers** also provide **lower-order functions**

High-order center (A):

- College, university
- Specialized clinics

Mid-order center (B):

- Secondary education, vocational school
- Hospital, specialist doctors

Low-order centers (C):

- Primary education
- Pharmacy, general practitioner, dentist

Christaller: central place theory

Formation of the Urban Hierarchy

Lower-order services appear in **higher-level centers** due to **agglomeration advantages**.

The **range of lower-order services** is smaller and does not cover the entire area — leading to the establishment of **new locations**.

Three organizing principles:

(a) Market Principle:

A **minimal number** of lower-order centers are established so that the **entire surface is covered**.

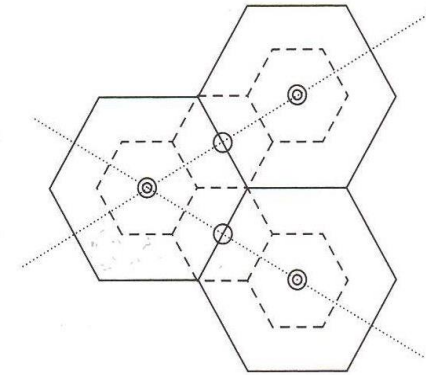
(b) Transportation Principle:

Lower-order centers are **optimally located** to **minimize transportation costs** from higher-order centers.

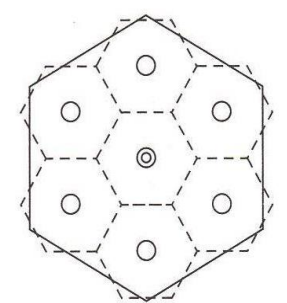
(c) Administrative Principle:

Each lower-order center is **clearly assigned** to a **specific higher-order center**.

(b) The transportation principle



(c) The administrative principle



4. Agglomeration economies

Definition and sources

Agglomeration Advantages:

- Firms become **more productive** thanks to agglomeration benefits.
- Growth is **faster in areas** where **positive externalities** are present.

Sources of agglomeration advantages (*Puga, 2010*):

- **Sharing:** use of labor and urban infrastructure
- **Matching:** meeting of supply and demand (e.g., employers and employees)
- **Learning:** knowledge spillovers and learning among firms

Static and dynamic agglomeration advantages:

- **Neo-Marshallian view:** local externalities improve firms' **static performance** (efficiency, cost)
- **Neo-Schumpeterian view:** local externalities enhance **innovative capacity**

Krugman: New Economic Geography

Explanation of Industry Concentration

- Increasing returns: stimulate agglomeration
- Transportation costs: encourage proximity to raw materials and intermediate goods
- Worker migration: increases labor supply and local market size



Paul Krugman (Princeton)
Nobel prize 2008

Krugman: centrum-periphery

Initially, equal distribution between the two regions: relocation of a firm has opposing effects

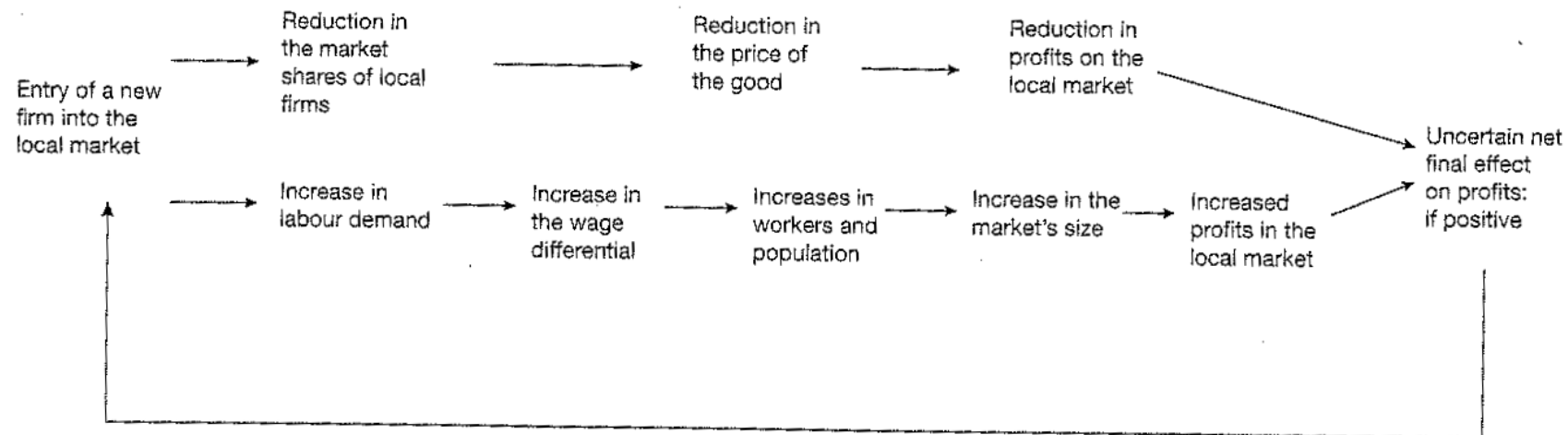
Competition effect:

- Depends on the **substitutability of products** (higher elasticity $\rightarrow$ greater profit loss)
- Also influenced by **transportation costs** (a decrease $\rightarrow$ lower prices $\rightarrow$ increased competition)

Demand effect:

- Driven by **increasing returns** (higher profit margins, stronger attraction for firms)
- Influenced by **income spent on products**

(a) The 'competition effect' and the 'demand effect'



High transportation costs:

- Stable equilibrium: firms do not relocate

Low transportation costs:

- Stable equilibrium: the industry concentrates in one region (where agglomeration advantages are greater)

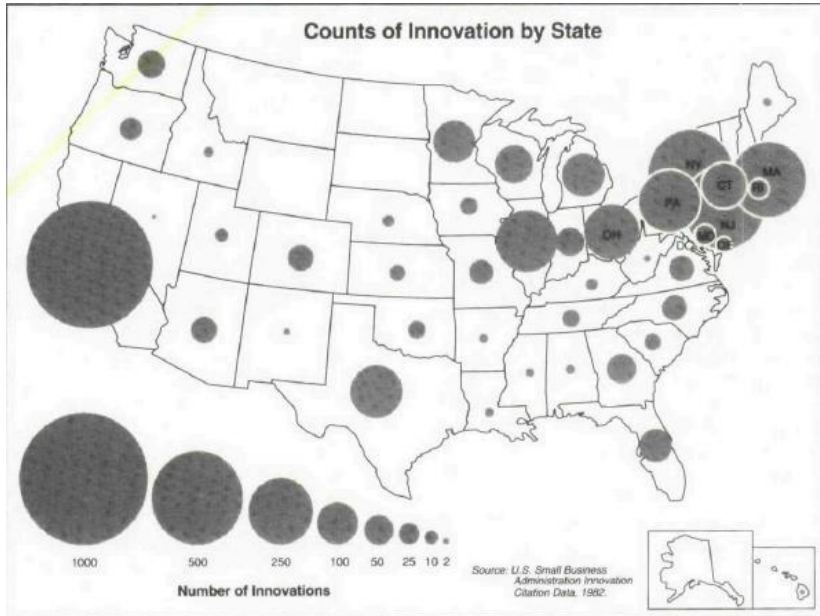
Moderately high transportation costs:

- Both equal distribution and core-periphery structure can be stable equilibria



5. Concentration of knowledge

Audretsch-Feldman 1996 – American Economic Review



- Concentration of technological innovation
- Spillovers are local
- Data: new products published in technological journals

- Gini index

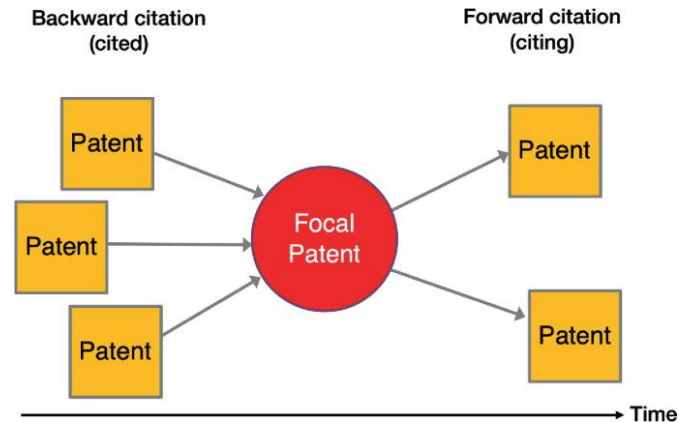
$$G = \frac{1}{2\bar{x}n^2} \sum_i \sum_j |x_i - x_j|$$

It compares the **average deviation** of an industry's share in a given region from its share in all other regions, **relative to the overall average**.

The larger the value, the more the industry is **concentrated in a few regions**.

- Comparison between production's and innovation's concentration
- Explanatory variables: natural resources, economies of scale, transportation costs, skilled workforce, R&D intensity, university R&D

Distance in knowledge spillovers



GEOGRAPHIC LOCALIZATION OF KNOWLEDGE SPILLOVERS AS EVIDENCED BY PATENT CITATIONS*

ADAM B. JAFFE
MANUEL TRAJTENBERG
REBECCA HENDERSON

We compare the geographic location of patent citations with that of the cited patents, as evidence of the extent to which knowledge spillovers are geographically localized. We find that citations to domestic patents are more likely to be domestic, and more likely to come from the same state and SMSA as the cited patents, compared with a “control frequency” reflecting the pre-existing concentration of related research activity. These effects are particularly significant at the local (SMSA) level. Localization fades over time, but only very slowly. There is no evidence that more “basic” inventions diffuse more rapidly than others.

The Quarterly Journal of Economics, Volume 108, Issue 3 (Aug., 1993), 577-598.



Journal of Urban Economics
Volume 42, Issue 3, November 1997, Pages 422-448



Regular Article

Local Geographic Spillovers between University Research and High Technology Innovations ☆

Luc Anselin ^a, Attila Varga ^{b c}, Zoltan Acs ^{d e}

Abstract

This paper re-examines the empirical evidence on the degree of spatial spillover between university research and high technology innovations. The familiar Griliches–Jaffe knowledge production function is estimated at both the state and the metropolitan statistical areas (MSA) level and extended with more precise measures of spatial spillover. Alternatives based on the gravity potential and covering indices are formulated for Jaffe's “geographical coincidence index” and found to provide strong evidence of local spillovers at the state level. At the MSA level, a distinction is made between research and development activities and university research in the MSA and in the surrounding counties. Evidence is found of local spatial externalities between university research and high technology innovative activity, both directly and indirectly via private research and development.

The gravity law

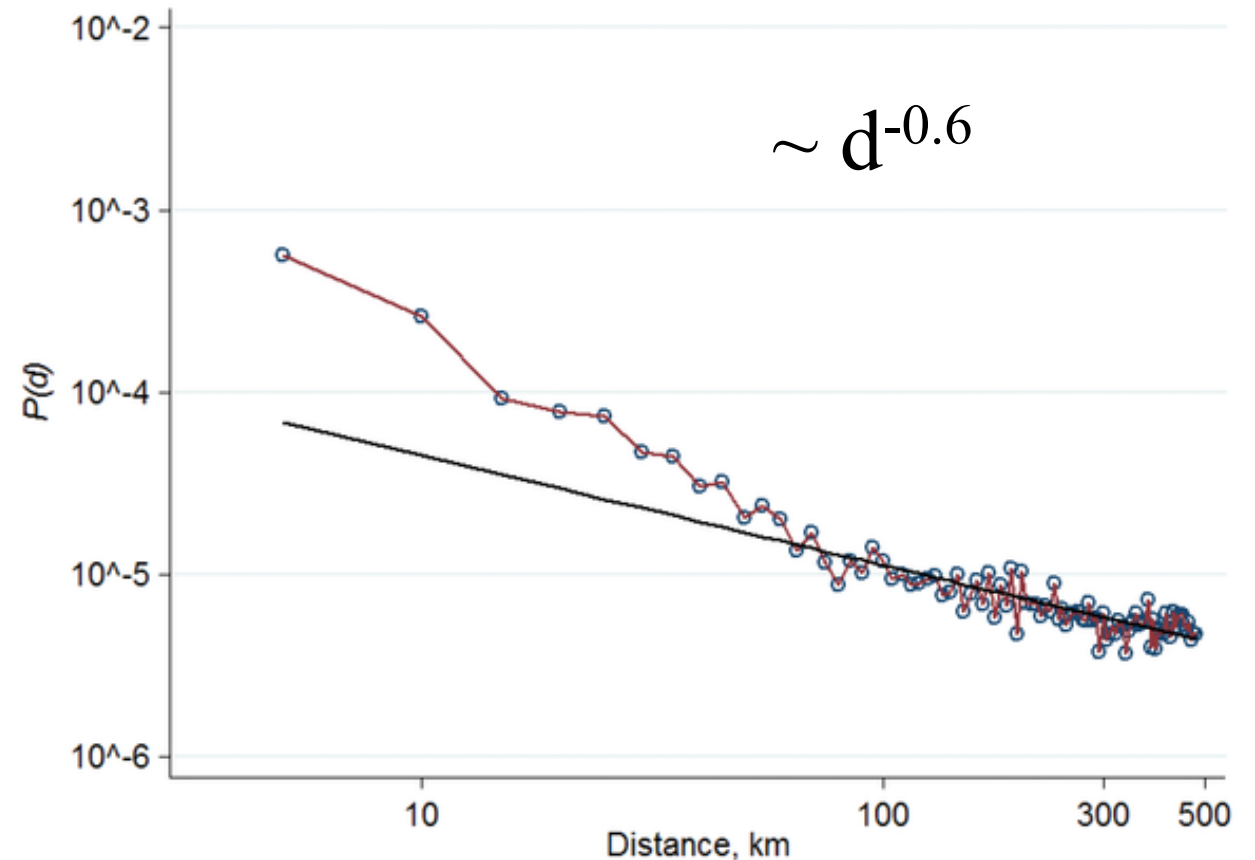
Zipf: $T_{ab} = K(P_a^\alpha P_b^\beta) / d_{ab}^\gamma$

Probability of relationships:

$N(d)$ – all possible connections

$L(d)$ – observed connections

$P(d) = L(d) / N(d)$



Urban scaling

$$Y(t) = Y_0 N(t)^\beta.$$

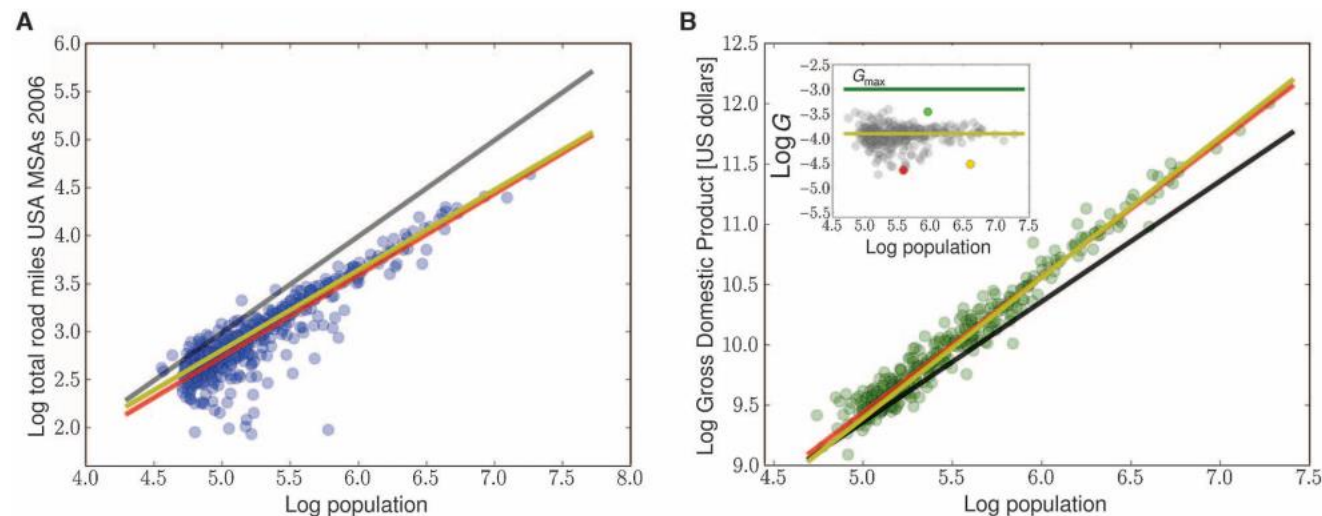


Fig. 1. Scaling of urban infrastructure and socioeconomic output. (A) Total lane miles (volume) of roads in U.S. metropolitan areas (MSAs) in 2006 (blue dots). Data for 415 urban areas were obtained from the Office of Highway Policy Information from the Federal Highway Administration (14). Lines show the best fit to a scaling relation $Y(N) = Y_0 N^\beta$ (red), with $\beta = 0.849 \pm 0.038$ [95% confidence interval (CI), $R^2 = 0.65$]; the theoretical prediction, $\beta = 5/6$ (yellow); and linear scaling $\beta = 1$ (black). **(B)** Gross metropolitan product of MSAs in 2006 (green dots). Data obtained for 363 MSAs from U.S. Bureau of Economic Analysis (14). Lines describe best fit (red) to data, $\beta = 1.126 \pm 0.023$ (95% CI, $R^2 = 0.96$); the theoretical prediction, $\beta = 7/6$ (yellow); and proportional scaling, $\beta = 1$ (black). The two best-fit parameters in each scaling

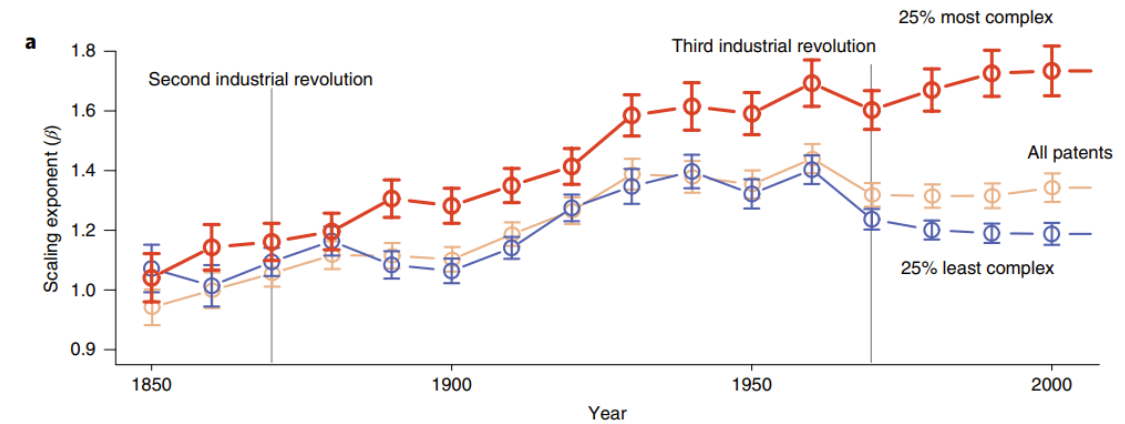
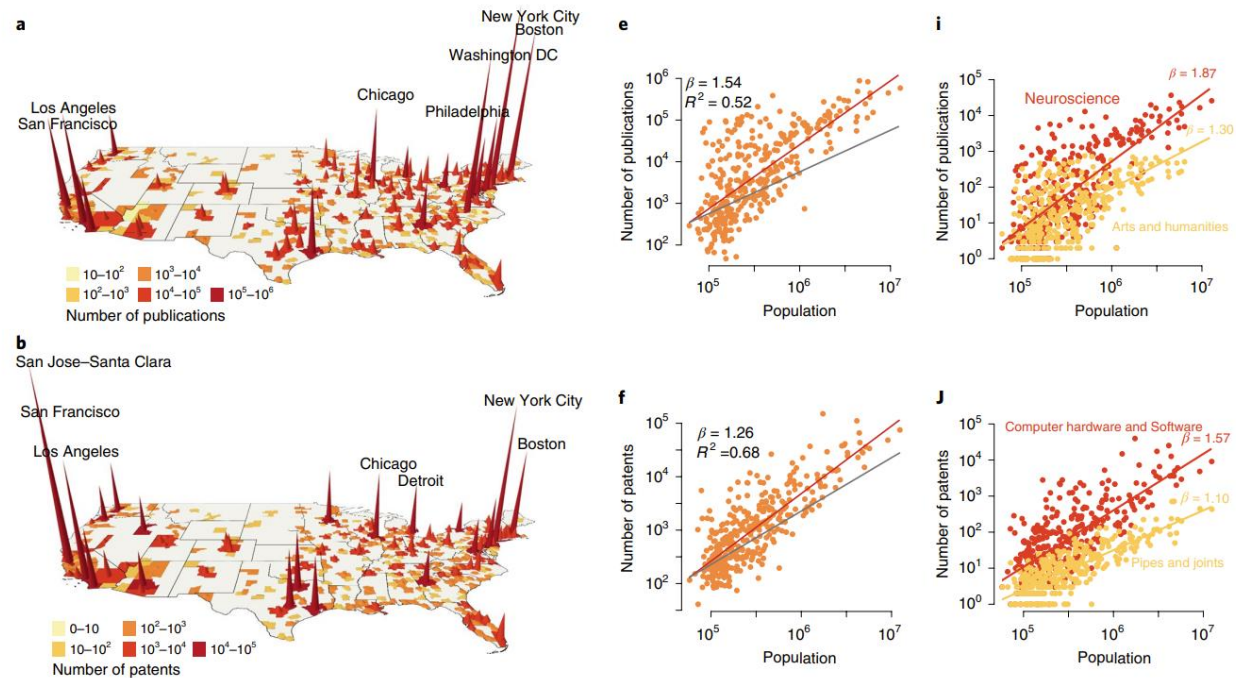
relation were estimated by means of ordinary least-squares minimization to the linear relation between logarithmically transformed variables (14). The inset shows the estimate of G for 313 U.S. MSAs and the conservation law $\frac{d \ln G}{d \ln N} = 0$ ($R^2 = 0.003$). G is measured as the product of gross domestic product and road volume, both per capita. As predicted by the theory, observed values of G for different cities cluster around its most likely value (mode, yellow line), which gives an estimate of the optimum G^* , and are bounded by the maximum $G_{\max} \approx 8G^*$ (green line); see also Fig. 2B. Several metropolitan areas, such as Bridgeport, Connecticut (green circle); Riverside, California (yellow circle); or Brownsville, Texas (red circle), are outliers, suggesting that they are suboptimal in terms of their transportation efficiency or amount of social mixing.

Table 1. Scaling exponents for urban indicators vs. city size

Y	β	95% CI	Adj- R^2	Observations
New patents	1.27	[1.25,1.29]	0.72	331
Inventors	1.25	[1.22,1.27]	0.76	331
Private R&D employment	1.34	[1.29,1.39]	0.92	266
"Supercreative" employment	1.15	[1.11,1.18]	0.89	287
R&D establishments	1.19	[1.14,1.22]	0.77	287
R&D employment	1.26	[1.18,1.43]	0.93	295
Total wages	1.12	[1.09,1.13]	0.96	361
Total bank deposits	1.08	[1.03,1.11]	0.91	267
GDP	1.15	[1.06,1.23]	0.96	295
GDP	1.26	[1.09,1.46]	0.64	196
GDP	1.13	[1.03,1.23]	0.94	37
Total electrical consumption	1.07	[1.03,1.11]	0.88	392
New AIDS cases	1.23	[1.18,1.29]	0.76	93
Serious crimes	1.16	[1.11,1.18]	0.89	287
Total housing	1.00	[0.99,1.01]	0.99	316
Total employment	1.01	[0.99,1.02]	0.98	331
Household electrical consumption	1.00	[0.94,1.06]	0.88	377
Household electrical consumption	1.05	[0.89,1.22]	0.91	295
Household water consumption	1.01	[0.89,1.11]	0.96	295
Gasoline stations	0.77	[0.74,0.81]	0.93	318
Gasoline sales	0.79	[0.73,0.80]	0.94	318
Length of electrical cables	0.87	[0.82,0.92]	0.75	380
Road surface	0.83	[0.74,0.92]	0.87	29

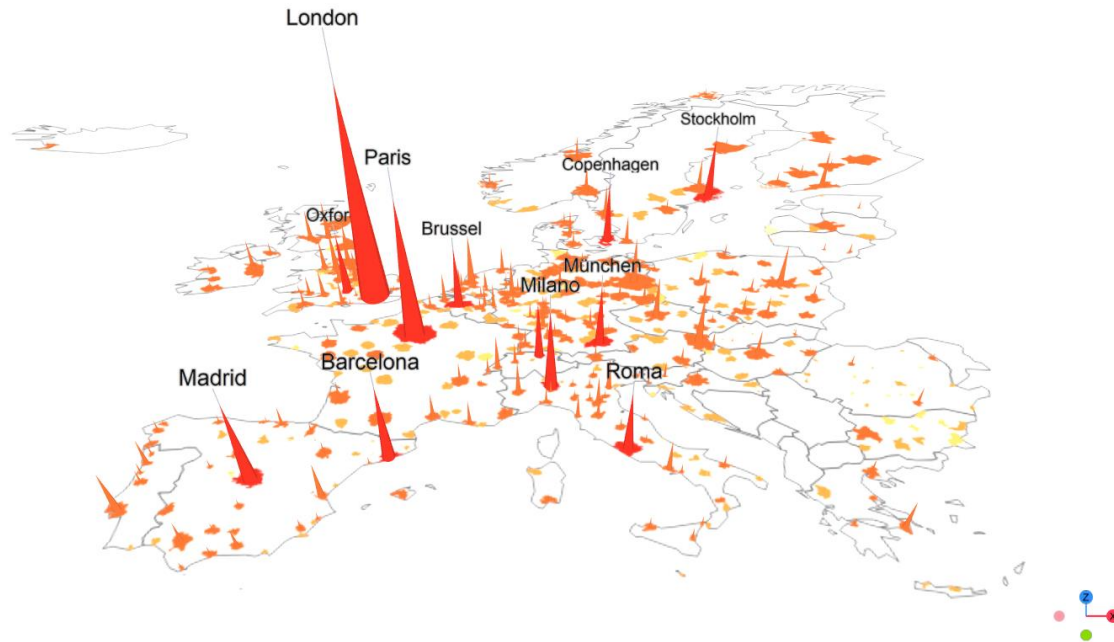
Complex economic activities concentrate in large cities

Pierre-Alexandre Balland ^{1,2*}, Cristian Jara-Figueroa ², Sergio G. Petralia ³, Mathieu P. A. Steijn ^{1,4}, David L. Rigby ⁵ and César A. Hidalgo ^{2,6,7,8}



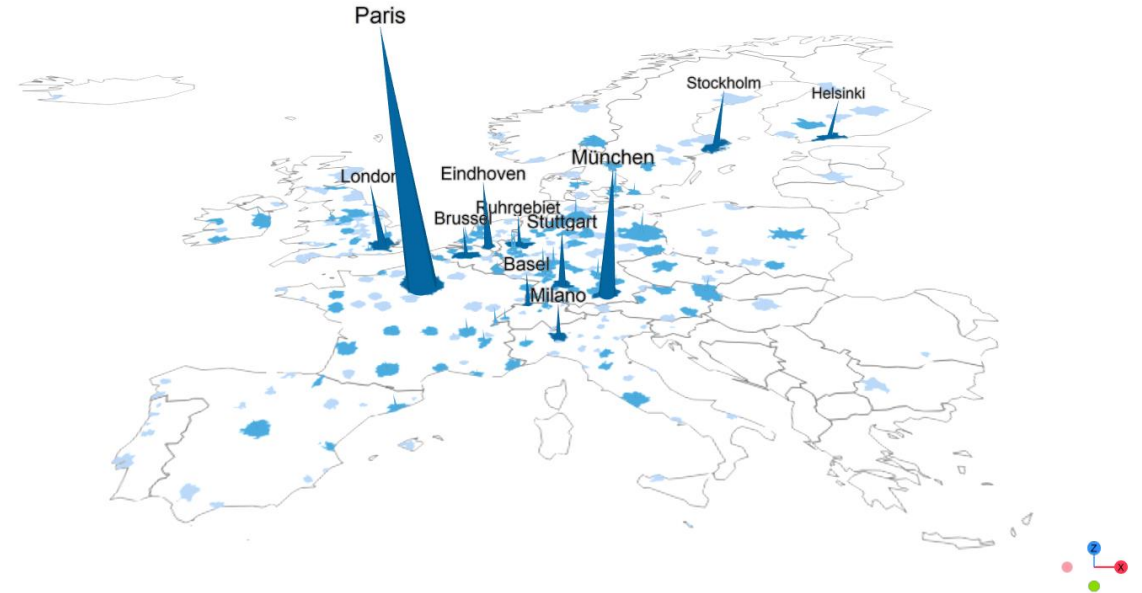
Geography of Knowledge Production in Europe, 2020

A. Scientific papers



Dominated by Paris, London, Madrid, Rome, Vienna — follows population.

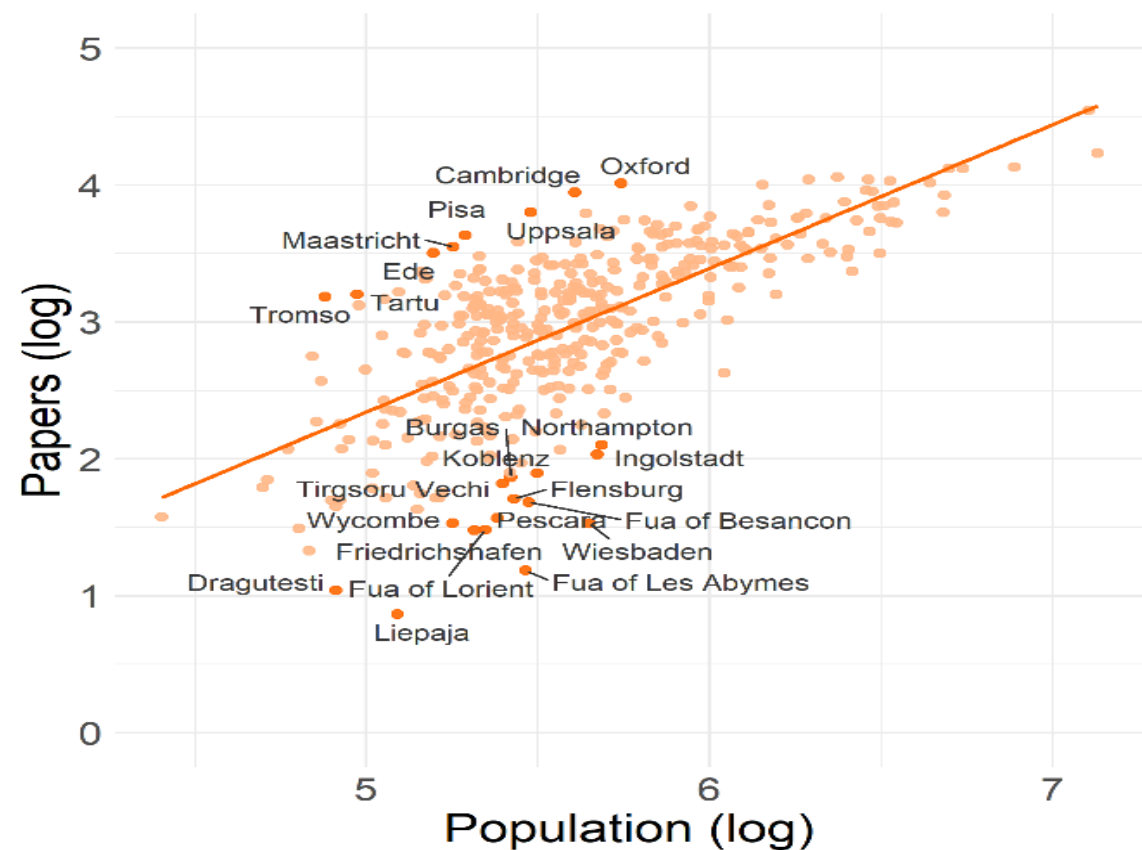
B. Patents



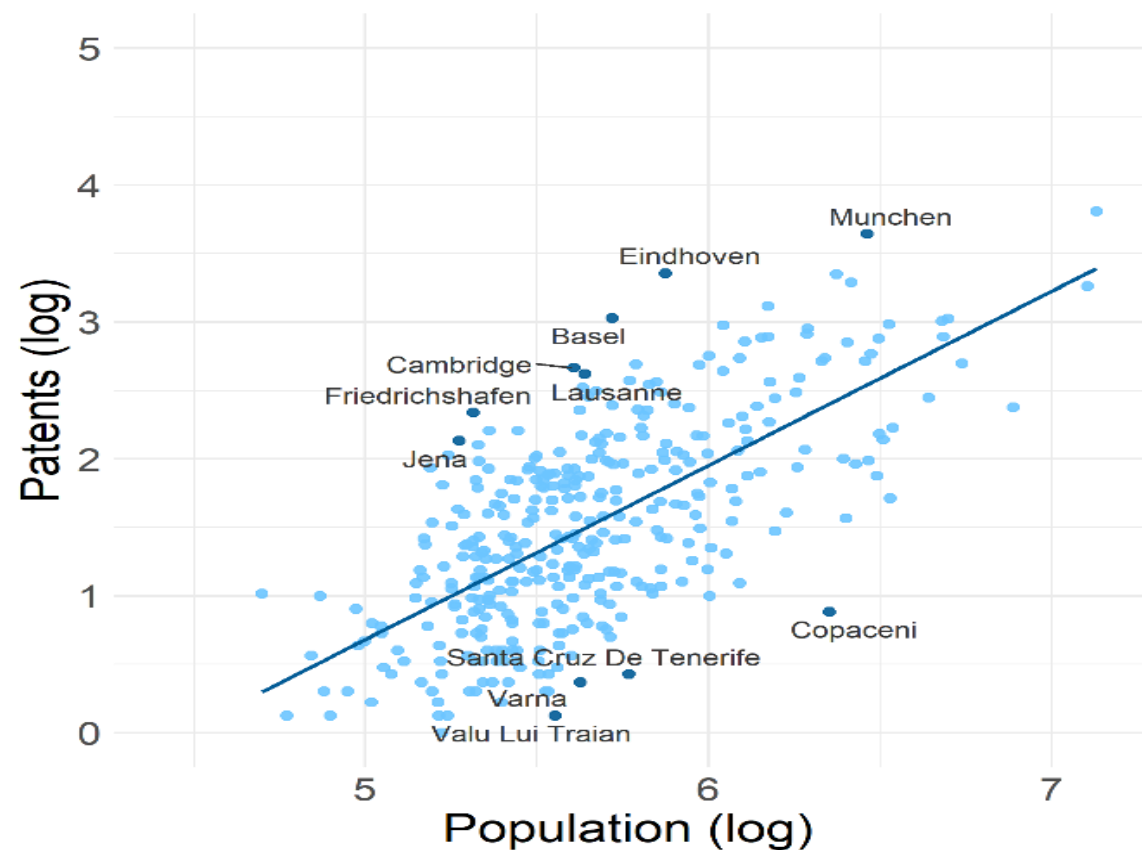
Paris, Munich, Eindhoven — concentrated in the 'blue banana' corridor.

Urban Scaling of Knowledge Production, 2020

C. Scientific papers ($\beta = 1.049$, $R^2 = 0.49$)

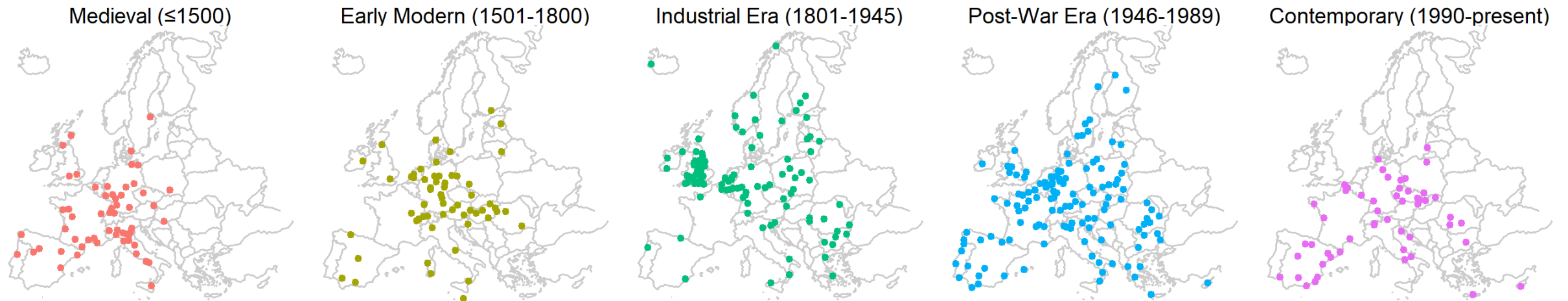


B. Patents ($\beta = 1.317$, $R^2 = 0.46$)



Papers scale nearly linearly. Patents scale superlinearly. Small/medium cities with old universities (Oxford, Cambridge, Tartu) punch above weight for papers; Eastern European cities underperform in patents.

Geography of University Founding Eras

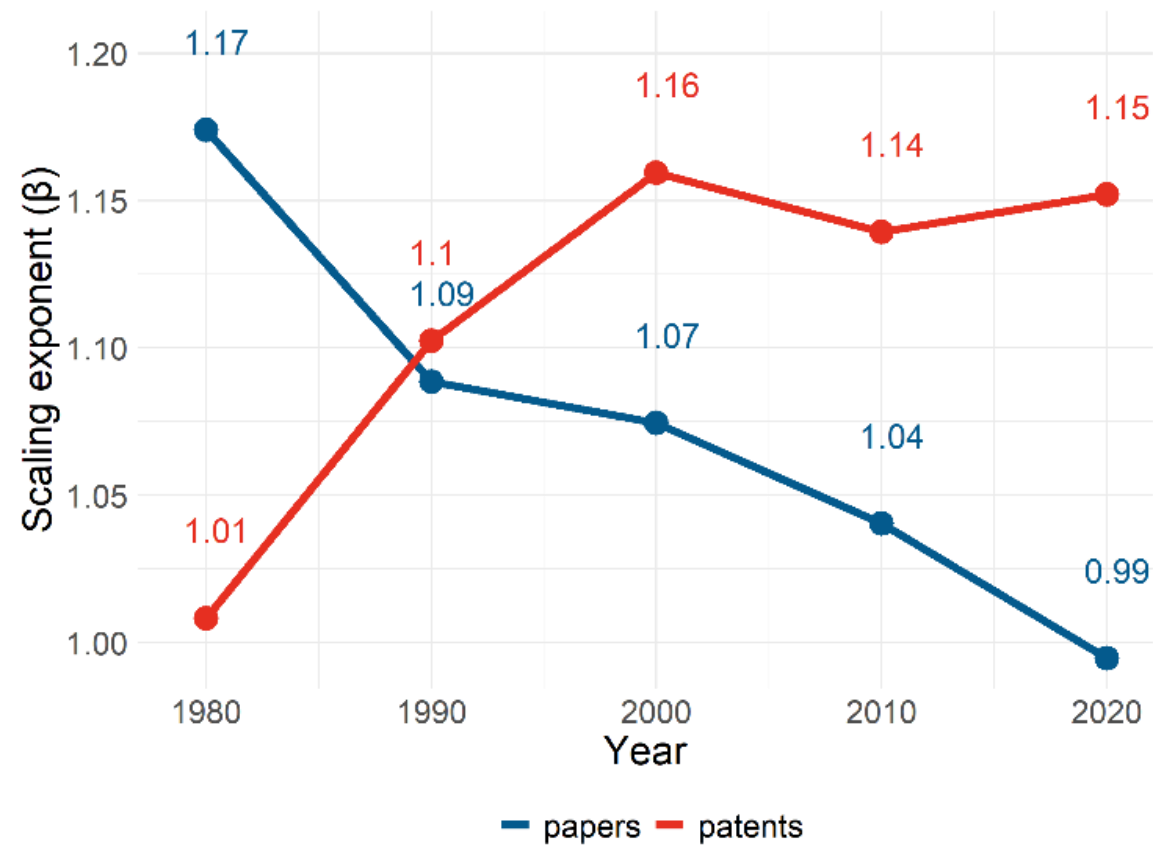


Medieval universities (≤1500) cluster in Western and Southern Europe. Industrial Era expansion moves to Britain and Germany. Post-War and Contemporary universities spread to Eastern Europe and the periphery.

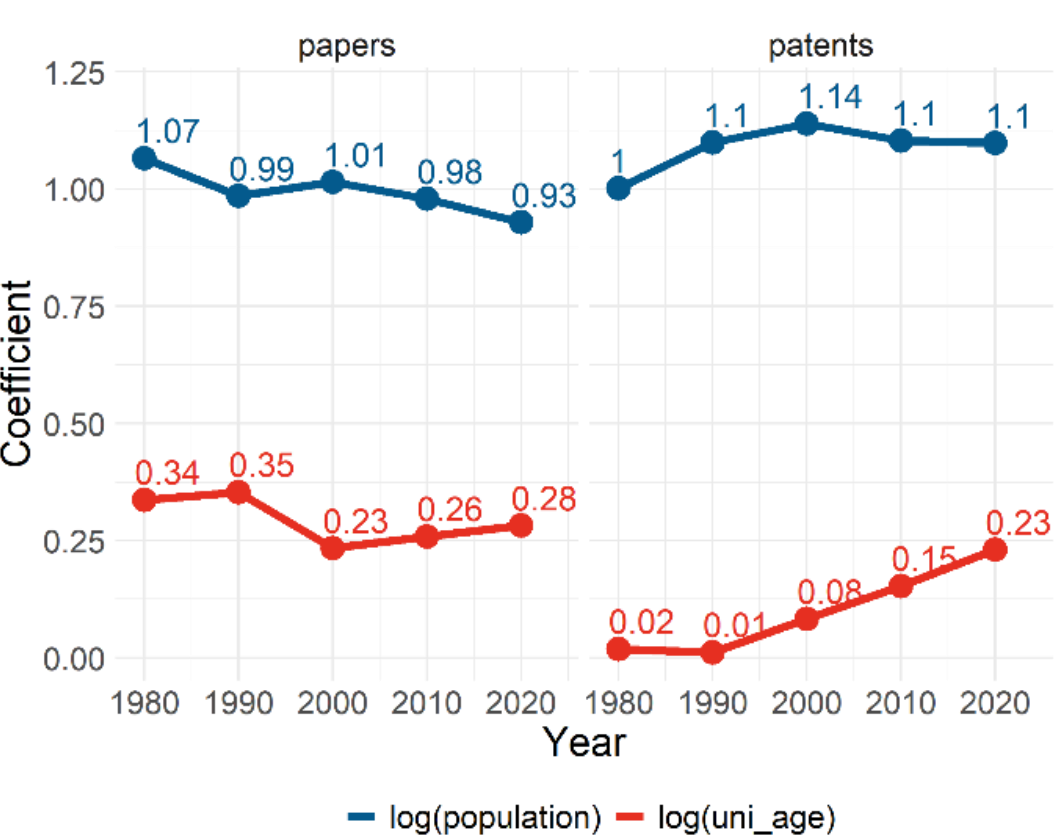
Source: EHESO database (European Commission, 2025). 408 FUAs classified by founding era of the oldest university in the city.

Urban Scaling Dynamics, 1980–2020 (113 cities with papers and patents in all years)

A. Scaling exponent β over time



B. Coefficients controlling for university age



Papers: β falls from 1.17 (1980) to 0.99 (2020) — deconcentration. Patents: β rises from 1.01 to 1.15 — growing concentration. University age effect on patents is rising (0.02→0.23) — history matters more over time.